Linear Algebra 2026 Homework 6

Due: November 6, 2026

Problems should be solved in English and calculation steps are required. For all T/F questions, you don't need to justify your answer.

All notations can be found in the book "Linear Algebra — the fourth edition" by Stephen H. Friedberg, Arnold J. Insel, Lawrence E. Spence. If you cannot locate them, feel free to contact the TAs.

Problem 1 [8 pts]. Label the following statements as true or false. Assume that all vector spaces are finite-dimensional.

(a) Every linear transformation is a linear functional.

(b) A linear functional defined on a field may be represented as a 1×1 matrix.

(c) Every vector space is isomorphic to its dual space.

(d) Every vector space is isomorphic to the dual of some other vector space.

(e) If T is an isomorphism from V onto V^* and β is a finite ordered basis for V, then $\mathsf{T}(\beta) = \beta^*$.

(f) If T is a linear transformation from V to W, then the domain of $(\mathsf{T}^t)^t$ is V^{**}.

(g) If V is isomorphic to W, then V^* is isomorphic to W^*.

(h) The derivative of a function may be considered as a linear functional on the vector space of differentiable functions.

Problem 2 [15 pts]. Define $f \in (\mathsf{R}^2)^*$ by $f(x, y) = 2x + y$ and $\mathsf{T} : \mathsf{R}^2 \to \mathsf{R}^2$ by $\mathsf{T}(x, y) = (3x + 2y, x)$.

(a) Compute $\mathsf{T}^t(f)$.

(b) Compute $[\mathsf{T}^t]_{\beta^*}$, where β is the standard ordered basis for R^2 and $\beta^* = \{f_1, f_2\}$ is the dual basis, by finding scalars a, b, c, d such that

$$\mathsf{T}^t(f_1) = af_1 + cf_2 \qquad \text{and} \qquad \mathsf{T}^t(f_2) = bf_1 + df_2.$$

(c) Compute $[\mathsf{T}]_\beta$ and $([\mathsf{T}]_\beta)^t$, and compare your results with (b).

Problem 3 [15 pts]. Let $\mathsf{V} = \mathsf{P}_1(\mathsf{R})$ and $\mathsf{W} = \mathsf{R}^2$ with respective standard ordered bases β and γ. Define $\mathsf{T} : \mathsf{V} \to \mathsf{W}$ by

$$\mathsf{T}(p(x)) = (p(0) - 2p(1),\, p(0) + p'(0)),$$

where $p'(x)$ is the derivative of $p(x)$.

(a) For $f \in \mathsf{W}^*$ defined by $f(a, b) = a - 2b$, compute $\mathsf{T}^t(f)$.

(b) Compute $[\mathsf{T}^t]_{\gamma^*}^{\beta^*}$ without appealing to Theorem 2.25.

(c) Compute $[\mathsf{T}]_\beta^\gamma$ and its transpose, and compare your results with (b).

Problem 4 [10 pts]. Prove that a function $\mathsf{T} : F^n \to F^m$ is linear if and only if there exist $f_1, f_2, \ldots, f_m \in (F^n)^*$ such that

$$\mathsf{T}(x) = (f_1(x), f_2(x), \ldots, f_m(x)) \qquad \text{for all } x \in F^n.$$

Hint: If T is linear, define $f_i(x) = (g_i\mathsf{T})(x)$ for $x \in F^n$; that is, $f_i = \mathsf{T}^t(g_i)$ for $1 \le i \le m$, where $\{g_1, g_2, \ldots, g_m\}$ is the dual basis of the standard ordered basis for F^m.

The following notation is used in Problems 5 and 6.

Definition 1. Let V be a finite-dimensional vector space over F. For every subset S of V, define the **annihilator** S^0 of S by

$$S^0 = \{f \in \mathsf{V}^* : f(x) = 0 \text{ for all } x \in S\}.$$

The canonical map $\psi : \mathsf{V} \to \mathsf{V}^{**}$ in Theorem 2.26 is defined by

$$\psi(x)(f) = f(x) \qquad (x \in \mathsf{V},\ f \in \mathsf{V}^*).$$

Problem 5 [25 pts].

(a) Prove that S^0 is a subspace of V^*.

(b) If W is a subspace of V and $x \notin \mathsf{W}$, prove that there exists $f \in \mathsf{W}^0$ such that $f(x) \neq 0$.

(c) Prove that $(S^0)^0 = \text{span}(\psi(S))$, where ψ is defined as in Theorem 2.26.

(d) For subspaces W_1 and W_2, prove that $\mathsf{W}_1 = \mathsf{W}_2$ if and only if $\mathsf{W}_1^0 = \mathsf{W}_2^0$.

(e) For subspaces W_1 and W_2, show that $(\mathsf{W}_1 + \mathsf{W}_2)^0 = \mathsf{W}_1^0 \cap \mathsf{W}_2^0$.

Problem 6 [10 pts]. Prove that if W is a subspace of V, then

$$\dim(\mathsf{W}) + \dim(\mathsf{W}^0) = \dim(\mathsf{V}).$$

Hint: Extend an ordered basis $\{x_1, x_2, \ldots, x_k\}$ of W to an ordered basis $\beta = \{x_1, x_2, \ldots, x_n\}$ of V. Let $\beta^* = \{f_1, f_2, \ldots, f_n\}$. Prove that $\{f_{k+1}, f_{k+2}, \ldots, f_n\}$ is a basis for W^0.

Problem 7 [9 pts]. Label the following statements as true or false.

(a) The rank of a matrix is equal to the number of its nonzero columns.

(b) The product of two matrices always has rank equal to the lesser of the ranks of the two matrices.

(c) The $m \times n$ zero matrix is the only $m \times n$ matrix having rank 0.

(d) Elementary row operations preserve rank.

(e) Elementary column operations do not necessarily preserve rank.

(f) The rank of a matrix is equal to the maximum number of linearly independent rows in the matrix.

(g) The inverse of a matrix can be computed exclusively by means of elementary row operations.

(h) The rank of an $n \times n$ matrix is at most n.

(i) An $n \times n$ matrix having rank n is invertible.

Problem 8 [15 pts]. Find the rank of the following matrices.

(a) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 4 \end{pmatrix}$

Problem 9 [10 pts]. For each of the following matrices, compute the rank and the inverse if it exists.

(a) $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 2 & 1 \\ -1 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}$

Problem 10 [10 pts]. For each of the following linear transformations T, determine whether T is invertible, and compute T^{-1} if it exists.

(a) $\mathsf{T} : \mathsf{P}_2(\mathsf{R}) \to \mathsf{P}_2(\mathsf{R})$ defined by $\mathsf{T}(f(x)) = (x+1)f'(x)$.

(b) $\mathsf{T} : \mathsf{R}^3 \to \mathsf{R}^3$ defined by

$$\mathsf{T}(a_1, a_2, a_3) = (a_1 + 2a_2 + a_3,\ -a_1 + a_2 + 2a_3,\ a_1 + a_3).$$

Problem 11 [15 pts]. Let $\mathsf{T}, \mathsf{U} : \mathsf{V} \to \mathsf{W}$ be linear transformations.

(a) Prove that $\mathsf{R}(\mathsf{T} + \mathsf{U}) \subseteq \mathsf{R}(\mathsf{T}) + \mathsf{R}(\mathsf{U})$. (See the definition of the sum of subsets of a vector space on page 22.)

(b) Prove that if W is finite-dimensional, then

$$\operatorname{rank}(\mathsf{T} + \mathsf{U}) \leq \operatorname{rank}(\mathsf{T}) + \operatorname{rank}(\mathsf{U}).$$

(c) Deduce from (b) that $\operatorname{rank}(A+B) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$ for any $m \times n$ matrices A and B.

Problem 12 [11 pts]. Prove that if B is a 3×1 matrix and C is a 1×3 matrix, then the 3×3 matrix BC has rank at most 1. Conversely, show that if A is any 3×3 matrix having rank 1, then there exist a 3×1 matrix B and a 1×3 matrix C such that $A = BC$.

Problem 13 [10 pts]. Let A be an $m \times n$ matrix with rank m. Prove that there exists an $n \times m$ matrix B such that $AB = \mathsf{I}_m$.

Problem 14 [10 pts]. Let $A, B \in \mathsf{M}_{n\times n}(\mathsf{R})$. Suppose $AB = BA = 0$ and $\operatorname{rank}(A^2) = \operatorname{rank}(A)$. Prove that

$$\operatorname{rank}(A+B) = \operatorname{rank}(A) + \operatorname{rank}(B).$$